

The Future of Mathematics:

From Problem-Solving to Theory-Development in the Age of AI

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ABSTRACT

Modern AI has crossed a historic threshold in mathematical problem-solving without thereby mastering theory-development. Problem-solving searches for a proof, algorithm, model, program, or construction inside a sufficiently specified world. Theory-development chooses or revises the objects, representations, definitions, questions, and standards of value that make such a world possible. My position is deliberately stronger than "AI is not there yet" but weaker than a claim of permanent human exceptionalism: scaling the present large-language-model recipe is unlikely, by itself, to produce reliable autonomous theory-development. Such systems may generate increasingly powerful theory-like moves, but a theory-developing AI would also need persistent research memory, tools for contact with mathematical and empirical consequences, long-horizon objectives, diverse hypothesis formation, and participation in expert processes that judge importance and fertility. Until then, the most credible future is AI as a formidable co-developer of theory, with human expertise still central to framing, abstraction, and judgment.

The central claim: present LLMs can increasingly solve within a mathematical world. Scaling them alone is unlikely to make them reliable autonomous authors of such worlds.

1 | The future of mathematics

This essay is a sequel to The Future of Programming. There I argued that AI increasingly compiles a formal statement of what a program should do into an implementation of how it does it. In mathematics, a theorem statement similarly specifies a problem and a proof implements the answer. But mathematics begins one level earlier than either analogy: someone must invent the language, select the question, and decide why an answer would matter (Rao, 2026b).

That earlier layer is the subject of this paper. AI can, in principle, reach it; nothing here depends on a mystical human faculty. My narrower claim is that today's neural systems are optimized for a different regime. Pretraining rewards prediction within records humans have already produced. Instruction tuning and preference optimization reward useful responses to goals that users supply. Formal proof systems and algorithm-search systems add exact evaluators. These ingredients are extraordinarily effective once the game is identifiable, but theory-development must sometimes change the game, the vocabulary, and the score (Brown et al., 2020; Ouyang et al., 2022).

William Thurston resisted a picture of mathematics as a pipeline of definitions, theorems, and proofs, arguing that mathematicians advance human understanding. Questions, representations, and usable ways of thinking are not preliminaries to the enterprise; they are much of it (Thurston, 1994). The practical issue for AI is therefore not simply whether it can produce a correct proof. It is whether it can originate and sustain a framework that makes many proofs worth seeking.

My proposed decomposition has two overlapping modes: problem-solving and theory-development. The boundary is not perfectly sharp, and advanced problem-solving often contains local acts of theory. Even so, the distinction identifies the missing capability more accurately than a contest between 'human creativity' and 'machine calculation.' It also suggests a concrete research agenda for systems that might eventually cross the gap.

2 | Two modes, one confusion

Problem-solving

In problem-solving, the mathematical world has largely been chosen in advance. The objects are named; the admissible operations are understood; a question is posed; and there is a reasonably crisp notion of success. The problem may be brutally difficult. It may require a construction nobody has seen, an ingenious invariant, or a proof that runs for hundreds of pages. Difficulty is not the distinguishing variable. Specification is.

A compact formalization is useful. Let L denote a language, A a set of accepted assumptions and methods, G a goal, and V a verifier or evaluative criterion. Problem-solving asks for an artifact p - a proof, algorithm, model, program, or construction - such that $V(p; L, A, G)$ accepts it or scores it highly. Olympiad mathematics is a clean example. So are much of algorithm design, quantitative modeling, and programming once the requirements have stabilized. The search space may be vast, but the game is identifiable.

Theory-development

Theory-development acts one level above this search. It chooses or revises L, A, G, and sometimes V. It asks what the right objects are, what should count as the same object, which features are essential, what a productive definition should preserve, which questions deserve attention, and what kind of result would explain rather than merely certify. A successful theory does not simply settle one goal. It creates a landscape in which many goals become visible, related, and tractable.

This work includes problem-posing, representation design, definition invention, analogy, abstraction, conjecture formation, the organization of lemmas into a reusable architecture, and judgments of mathematical taste. Its outputs are often evaluated only retrospectively. A definition that initially looks artificial may become indispensable because it supports decades of theorems. A theorem may be correct yet sterile. A technically weaker result may matter more because it reveals the right coordinate system. Theory-development is therefore not just harder search. It partly constructs the search space and the value function.

Dimension	Problem-solving	Theory-development
Objective	Supplied or tightly bounded	Chosen, reframed, or created
Vocabulary	Mostly fixed	Invented, transferred, or reorganized
Evaluation	Correctness or score is available	Interest and fertility are partly endogenous
Typical output	Answer, proof, algorithm, model	Framework, abstraction, research program
Failure mode	Incorrect or suboptimal result	Correct but unilluminating or sterile theory
Time horizon	Usually bounded	Open-ended and socially validated

Table 1. The distinction is about who specifies the mathematical world, not about how difficult the resulting theorem is.

A reinforcement-learning analogy

For someone trained in reinforcement learning, the distinction can be stated provocatively. Problem-solving resembles optimizing a policy inside a given Markov decision process. The states, actions, dynamics, and reward may be complex or only partially known, but they define the task. Theory-development resembles deciding what the state should be, which distinctions can be discarded, what actions are meaningful, where the boundary of the environment lies, and what reward captures the

purpose. In real work, model and policy are developed together. But a breakthrough in policy optimization does not by itself show mastery of model or objective design.

This analogy also explains why the distinction is not metaphysical. Humans do not possess a magic faculty called theory. We perform a collection of meta-level activities for which present AI systems have weak or indirect training signals. Those signals can improve. The claim of this paper is empirical and architectural, not a proof of permanent human exceptionalism.

3 | What a varied mathematical life taught me

I offer my experience here cautiously. I am not an elite mathematician, nor have I moved a field of pure mathematics forward. My useful vantage point is range: olympiad problems, algorithms and abstract algebra, finance and retail models, software, reinforcement learning, exposition, and teaching. Moving among these settings made the difference between the two modes unusually visible. It is not a distinction between pure and applied work, or academia and industry. It is a distinction between operating inside a mathematical world and helping to construct one (Rao and Jelvis, 2022; Rao, 2026a).

Problem-solving becomes legible to a machine

In olympiad work, the problem arrives as a small, finished universe. Professional modeling often reaches the same condition after the difficult business conversation is over: price this security under these assumptions; allocate this inventory subject to these constraints; implement an algorithm that meets these requirements. The solution may demand search, decomposition, symbolic control, local ingenuity, coding discipline, and error correction. These are serious forms of intelligence. But the destination is externally legible, so attempts can be compared and corrected.

Many Colab notebooks on my Stanford page fit this description: retirement planning, inventory control, clearance pricing, price elasticity, order placement, and cricket strategy. I am proud of the care that went into them, but I do not regard them as protected human territory. Given their objectives, assumptions, and data, a strong LLM-assisted system can now reproduce a meaningful share of that work faster than I could, and in some cases improve it through broader search, cleaner code, and more exhaustive testing. My own past problem-solving is part of the evidence that AI's advance is real, not merely a benchmark curiosity (Rao, 2026a).

Theory-development starts before the calculation

The experience feels different when the work begins without an agreed representation. In The Newsvendor meets the Options Trader, a derivatives payoff and a retail inventory cost use different professional vocabularies but share the same mathematical form. Representing underage and overage costs as weighted option payoffs allows put-call parity, moneyness, delta, and market neutrality to acquire inventory interpretations (Rao, n.d.). Once the bridge is proposed, an AI can verify the identity, test distributions, derive sensitivities, and generate variants. The prior act is deciding that this bridge is meaningful rather than one of countless superficial analogies.

That judgment grew out of contact with both domains. The task was not to retrieve a formula from Finance and insert it into Retail. It was to decide what details could be forgotten, which structure should be preserved, and whether the resulting translation would make useful questions travel in both directions. After that representational choice, much of the remaining work becomes problem-solving again.

The same pattern appeared when financial decisions I had known in practice were recast as Markov decision processes for teaching and for Foundations of Reinforcement Learning with Applications in Finance. Portfolio choice, hedging, execution, market making, inventory, and clearance pricing could then share the language of states, actions, rewards, values, and Bellman operators (Rao and Jelvis, 2022). An AI given that ontology can derive equations and implementations. The theory-developing step is choosing a state that forgets enough but not too much, a reward that represents the real purpose, and a common frame that earns its complexity by enabling transfer.

Function Approximations as Affine Spaces makes a smaller but revealing representational move. Instead of treating a function approximator merely as a parameterized formula, it treats its parameter domain as an affine space, gradients as vectors of translation, and stochastic gradient descent as movement in that space (Rao and Jelvis, 2022, Appendix F). The algorithm does not change; the ontology does. AI can check the definitions, map them into code, and extend examples once this view is supplied. It is less reliable at deciding, without an expert agenda, that this is the abstraction worth building and at judging whether its conceptual economy survives across uses.

Teaching reveals another kind of expertise

My Abstract Algebra and Category Theory notes are teaching materials, not original contributions. Yet teaching theories exposes a fact that finished textbooks can conceal. An LLM can state a definition, prove an exercise, and generate many examples. The teacher's harder task is to choose the sequence of definitions, examples, non-examples, and diagrams through which an abstraction becomes a way of seeing. A formally correct presentation can still be conceptually dead (Rao, 2016; Rao, 2017).

This is also why simply training on the mathematical literature may be insufficient. Published mathematics records the successful route far better than the abandoned ontologies, misleading analogies, tacit judgments, and years of experience that selected it. A model can imitate the destination without learning the full process by which a community decided that the destination was worth reaching.

The lesson is about priors, not mystique

Across these examples, problem-solving relies on powerful search within a comparatively stable frame. Theory-development relies more heavily on representational judgment: what to preserve, what to ignore, which analogy is structural, which question is fertile, and when a failed path should alter the frame itself. Breadth of experience supplies imperfect but useful priors about those choices. This is not evidence of a non-computational soul. It is evidence that the relevant intelligence is trained by sustained interaction with domains, consequences, learners, and other experts - sources of feedback that present LLM training captures only indirectly.

4 | What the strongest AI results establish

The evidence for AI's mathematical power no longer requires a long recital. In 2025, an advanced Gemini system produced natural-language solutions at an officially certified IMO gold-medal level. Olympiad problem-solving is no longer a plausible boundary of uniquely human ability (Google DeepMind, 2025b).

More importantly, AlphaProof Nexus has proved previously open problems, including Erdos and OEIS conjectures (Tsoukalas et al., 2026). That is research-level mathematics. Yet its operating frame is revealing: the agent receives a Lean theorem, definitions, imports, and a proof hole; Lean supplies an exact verifier. It may invent helper lemmas and local definitions, but the overarching goal and formal world are supplied. Calling this frontier problem-solving is not to diminish it. It identifies where the autonomy resides.

FunSearch and AlphaEvolve produce genuine novelty in algorithms and constructions, sometimes improving the best known human result. They are most effective when progress can be scored automatically: an evaluator, and often a program skeleton, turns discovery into a search over candidates (Romera-Paredes et al., 2024; Google DeepMind, 2025a). These systems show that difficult, open, and novel do not automatically mean theory-developing. A result can be all three while the goal, representation, and measure of success remain human-specified.

The boundary is porous. In knot theory and representation theory, machine learning exposed patterns in large datasets; mathematicians selected promising relationships, introduced a new quantity, formulated

conjectures, and proved results. The work was an iterative human-machine loop, not a simple division between human creativity and machine calculation (Davies et al., 2021).

The current record therefore supports a more precise conclusion than either triumphalism or dismissal. AI performs increasingly deep search, decomposition, code generation, formalization, and local concept invention. It can contribute to theory-development when embedded in a human research process. What has not been demonstrated reliably is autonomous choice and revision of the problem, the ontology, the evaluator, and the long-term research program.

5 | Can AI become a developer of theory?

My answer has three layers. Today's LLMs are not reliable autonomous theory-developers. Scaling the current recipe may produce increasingly impressive theory-like acts, but is unlikely by itself to close the gap, because the missing element is not only reasoning power; it is the formation and revision of goals, representations, evaluators, and research programs. AI in a broader sense could cross that frontier, but it would have to become a different kind of compound research system. This is a forecast about the route, not a theorem about an ultimate limit.

The training signal points toward answers, not origins

An autoregressive language model learns to predict continuations of records produced by people. Post-training then teaches it to respond helpfully to instructions and human preferences (Brown et al., 2020; Ouyang et al., 2022). This can create rich internal representations and surprising generalization; it should not be dismissed as retrieval. But the objective primarily rewards a plausible or preferred continuation after a context and request have been supplied. It does not directly reward the decades-long downstream fertility of a new definition, or the wisdom of refusing the question in the prompt and replacing it with a better one.

The mathematical corpus has a second bias. It contains polished theories, successful proofs, and retrospective explanations in much greater proportion than failed ontologies, abandoned notations, unproductive questions, and the tacit reasons experts changed direction. The training data therefore overrepresent what theory looks like after selection. More data about finished mathematics can improve imitation of theory's surface while leaving the selection process only indirectly observed.

The goal and the evaluator are still supplied

Most present systems begin after crucial theoretical decisions have been made. A user writes the prompt; a formalizer chooses the theorem and imports; a benchmark fixes the answer; an evaluator scores the

program. Agentic loops can break a goal into subgoals, call tools, remember intermediate results, and propose local definitions. These are real advances. Yet they generally optimize within a task whose purpose and success signal come from outside the agent.

Theory-development is awkward for this architecture because its value function is partly endogenous. A theorem is either proved or not, but a theory can be internally correct and still sterile. Its value may lie in explanatory compression, transfer, extensibility, pedagogy, or questions that become visible only years later. The decisive evaluator may be the hundredth result enabled by a concept, not the first output that mentions it. Current mathematical AI advances fastest precisely where this delayed judgment can be replaced by an exact verifier or a numerical score.

The representation must sometimes change

Abstraction is disciplined forgetting. A state representation excludes most facts about the world; a group forgets what its elements are made of; an applied model discards operational detail so that a decision becomes tractable. The difficult act is deciding what may be removed without destroying the phenomenon of interest. In my finance-to-retail work, the meaningful bridge was not the most verbally similar analogy but the one that preserved payoff structure and allowed useful concepts to transfer. A language model can generate many analogies. Theory requires selecting, testing, and sometimes living with one long enough to discover whether it reorganizes the domain.

This also clarifies why a model's invention of a lemma or definition is not irrelevant to theory-development. It is a point on the continuum. A temporary definition that closes a supplied proof goal is a local theory-like act. A concept that changes the natural questions, survives across examples, and supports results no one initially targeted is a stronger one. Current systems demonstrate the first and occasionally glimpse the second. The issue is sustained scope, not a protected label.

Novelty is not yet a research program

Theory-development unfolds through failure. A conjecture collapses, a definition becomes too broad, an example once treated as central is reclassified as noise, or a proof obstruction reveals the right invariant. The researcher must preserve a coherent account of why the program matters while allowing that account to change. A long context window or external memory can store this history. It does not by itself supply a persistent epistemic life: an evolving model of the domain, its unresolved tensions, the reliability of its tools, and the reasons for continuing or abandoning a line of thought.

Nor is theory-development purely individual. Mathematics is a living system of attention, notation, explanation, standards, and trust. Researchers acquire taste through examples, mentors, failures, and argument with other researchers. Machines may eventually learn analogous judgments; there is no principled reason to reserve them for biology. But the feedback available to today's systems is dense for correctness and short-run preference, sparse for importance and long-run fertility. Recent critiques of autonomous AI science identify the same general obstacles: externally supplied problem selection, missing records of failure and tacit knowledge, short-horizon benchmarks, and weak contact with experimental feedback (Bisht et al., 2026).

The strongest objection is scale

The strongest objection to this position is that scale has repeatedly turned apparent architectural limitations into emergent capability. Next-token prediction learned translation, coding, proof tactics, and forms of abstraction without separate modules for each. A sufficiently capable model might also infer mathematical taste from the literature, generate its own questions, and use tools and simulated communities to evaluate them. I take this objection seriously. Any categorical claim that LLMs can never develop theory would be historically reckless.

But emergence does not make every extrapolation equally warranted. Fluency at proposing definitions can be measured immediately; fertility cannot. Solving a theorem selected by a human supplies a dense terminal condition; deciding which theorem deserves years supplies none. A model may become better at each local component while remaining dependent on a human-maintained agenda. The evidence needed to cross that inference is not another medal or even another open proof. It is repeated, minimally scaffolded construction of frameworks that experts adopt and use for results the system did not target.

What would have to change

A credible theory-developing AI would therefore be more than a larger conversational model. It would need a persistent and revisable world model of a research domain; formal, computational, simulation, and empirical tools that expose consequences; memory that preserves failed paths and the reasons they failed; mechanisms that maintain genuinely diverse hypotheses rather than converging too quickly on conventional continuations; and long-horizon processes for choosing which anomalies and analogies deserve attention.

It would also need evaluative participation. Expert mathematicians and domain practitioners would initially supply criticism, historical context, and judgments of usefulness. Over time, a system might internalize more of that role by predicting downstream theorem yield, transfer, compression, and adoption, while remaining able to revise those proxies when they distort the work. In applied theory, it would need contact with the world that the model is meant to organize, not only text about that world.

This architecture is possible in principle, and fragments of it already exist. But describing it makes clear why scaling LLM inference alone is not the same research program. The likely path is a coupled system in which neural models supply broad generative intelligence, symbolic tools enforce exactness, persistent agents maintain evolving programs, experiments provide resistance from the world, and expert communities shape and eventually share the judgment of value. For the foreseeable future, human expertise in theory-development is not merely ceremonial supervision; it is part of the mechanism that makes the work theory rather than prolific output.

6 | From a binary to a specification frontier

The two-part classification is most useful when refined into a frontier. Mathematical tasks vary along at least three axes: who selects the goal, who supplies the representation, and who supplies the evaluator. A closed task fixes all three. A more autonomous task allows the system to revise one or more. Theory-development lies toward the open end, where goals, representations, and standards co-evolve.

Level	Goal	Representation	Evaluation
1	Fixed exercise	Fixed	Exact verifier
2	Fixed theorem	Mostly fixed; lemmas open	Formal or expert check
3	Open problem	Fixed domain; methods open	Target result specified
4	Research direction	Definitions may change	Mixed and delayed
5	Field construction	Ontology is invented	Fertility judged over time

Table 2. A proposed specification frontier. Current systems are strong at Levels 1-2 and increasingly consequential at Level 3; human-AI collaborations are beginning to explore Level 4.

This framing avoids two errors. It does not reserve 'real mathematics' for humans whenever AI succeeds. Nor does it confuse movement on one axis with completion of the whole journey. AlphaProof Nexus at an open Erdos problem is Level 3 even if the proof is profound. An AI-human system that discovers a useful invariant and reorganizes a research agenda may be entering Level 4 even if its first theorem is elementary.

7 | What would count as AI theory-development?

If the thesis is scientific rather than protective, it needs falsification criteria. I would substantially revise my position if an AI system repeatedly demonstrated the following behaviors with limited human scaffolding.

Independent problem choice. From a broad and messy literature, the system identifies neglected questions that experts later judge important, rather than selecting from a prepared benchmark.

Definition invention. It introduces concepts not merely as proof conveniences but because they compress examples, expose invariants, and support unexpected theorems.

Cross-domain transfer. It recognizes a structural equivalence between fields whose surface vocabularies and datasets differ, then uses the bridge to generate new results on both sides.

Research-program persistence. It sustains and revises a line of inquiry over months, learns from failed conjectures, and maintains a coherent account of why the program matters.

Explanatory and downstream fertility. Its representations change how experts think, and independent researchers use them to obtain results the system did not target or foresee.

These criteria are intentionally harder to benchmark than theorem accuracy. That difficulty is part of the point. We have optimized AI for environments with cheap feedback because those environments support scaling. Theory-development asks us to build systems that operate when feedback is delayed, contested, and partly produced by the work itself. A benchmark may need longitudinal expert panels, blind comparison of alternative conceptual frameworks, and measurement of downstream reuse rather than one-shot correctness.

8 | AI as a co-developer of theory

Autonomy is not the only measure of consequence. AI can transform theory-development while remaining inside a human-led research loop. It can map literatures, propose analogies, enumerate examples and counterexamples, search for invariants, formalize conjectures, verify proofs, compare alternative definitions, and test whether a proposed abstraction actually compresses the cases it claims to unify. By shortening the cycle between an idea and its consequences, it can improve human judgment as well as execution.

My own examples suggest a practical division. Once the Newsvendor-options bridge is proposed, AI can derive the identity, stress it under distributions, search for extensions, and formalize the sensitivities. Once function approximations are represented as affine spaces, it can test whether the vocabulary handles linear models and neural networks consistently, connect the definitions to code, and expose edge cases. In teaching Category Theory or Abstract Algebra, it can generate exercises and multiple explanations. The human contribution remains the evolving judgment that this bridge, ontology, or pedagogical sequence is the one worth pursuing.

That division will not stay fixed. AI will absorb progressively more local theory-like acts, while people learn to collaborate with it at higher levels. The relevant unit may become the human-AI research system rather than either participant. Work in knot theory and representation theory already illustrates this loop: machine pattern recognition makes structure visible, human judgment converts selected patterns into concepts and conjectures, and formal reasoning tests the resulting account (Davies et al., 2021).

For education, the implication is uncomfortable but constructive. Training students primarily to outperform machines on closed-form exercises is no longer defensible. Students still need technique; without it, they cannot recognize a bad abstraction or inspect an AI proof. But curricula should put more weight on problem formulation, examples and counterexamples, definition design, model criticism, cross-domain analogy, exposition, and the reconstruction of why a theorem matters. The goal is not to preserve human jobs by retreating to vagueness. It is to teach the part of mathematics that our exams have historically under-measured.

For industry, the same principle applies. Once an optimization problem is faithfully specified, AI can increasingly build and solve the machinery. The largest failures often occur earlier: optimizing the wrong objective, omitting a critical state variable, treating an organizational constraint as a law of nature, or confusing a measurable proxy with enterprise value. In finance, retail, and supply-chain planning, the mathematics of choosing the model can dominate the mathematics of solving it. AI should be used aggressively to challenge a model and explore its implications, without confusing fluent implementation with faithful specification.

9 | Conclusion

The sequel from programming to mathematics is now clear. In programming, a human increasingly specifies what a program should do and AI compiles an implementation. In problem-solving mathematics, a human supplies a theorem or model and AI increasingly finds the proof, algorithm, or code. Theory-development begins one level earlier: it invents and revises the specification, the language in which it can be expressed, and the reason to care.

Today's LLMs are not designed around that long-horizon, self-revising task. Scaling them will make them better contributors and may yield genuine conceptual surprises, but reliable autonomous theory-development is unlikely to emerge from scale alone. It requires persistent research programs, resistance from formal and empirical worlds, diversity of hypotheses, and evaluators that can learn what explanatory and downstream value mean. Those capabilities may be built; the claim is not that they are forbidden.

Until they are demonstrated, the sensible position is neither complacency nor surrender. AI already surpasses much human problem-solving and should be welcomed as a co-developer of theory. Human mathematicians and domain experts remain essential where the work is to choose what to ignore, which analogy deserves belief, what question is worth asking, and when a framework has become a new way of seeing.

The decisive milestone will not be another solved problem. It will be an AI that helps create the mathematical world in which previously unseen problems become natural.

References

- Bisht, A., et al. (2026). Agentic AI Scientists Are Not Built For Autonomous Scientific Discovery. arXiv:2605.08956. [arXiv](#)
- Brown, T. B., et al. (2020). Language Models are Few-Shot Learners. Advances in Neural Information Processing Systems, 33. [arXiv](#)
- Davies, A., Velickovic, P., et al. (2021). Advancing mathematics by guiding human intuition with AI. Nature, 600, 70-74. [Nature](#)
- Google DeepMind. (2025a). AlphaEvolve: A Gemini-powered coding agent for designing advanced algorithms. [Google DeepMind](#)
- Google DeepMind. (2025b). Advanced version of Gemini with Deep Think officially achieves gold-medal standard at the International Mathematical Olympiad. [Google DeepMind](#)
- Ouyang, L., et al. (2022). Training language models to follow instructions with human feedback. Advances in Neural Information Processing Systems, 35. [arXiv](#)
- Rao, A. (n.d.). The Newsvendor meets the Options Trader. [Source and PDF](#)
- Rao, A. (2016). Abstract Algebra in 3 Hours. [Slide deck](#)
- Rao, A. (2017). Category Theory Made Easy with (Ugly) Pictures. [Slide deck](#)
- Rao, A. (2026a). Papers, slides, articles, and Colab notebooks. [Stanford work archive](#)
- Rao, A. (2026b). The Future of Programming: From Problem Specifications to AI-Compiled Implementations - A Type-Theoretic Foundation. [Source and PDF](#)
- Rao, A., and Jelvis, T. (2022). Foundations of Reinforcement Learning with Applications in Finance. CRC Press. [Open-access author PDF](#)
- Romera-Paredes, B., et al. (2024). Mathematical discoveries from program search with large language models. Nature, 625, 468-475. [Nature](#)
- Thurston, W. P. (1994). On proof and progress in mathematics. Bulletin of the American Mathematical Society, 30(2), 161-177. [arXiv](#)
- Tsoukalas, G., Kovsharov, A., Shirobokov, S., et al. (2026). Advancing mathematics research with AI-driven formal proof search. arXiv:2605.22763. [arXiv](#)

Author's note. *This position paper's taxonomy and forecasts are arguments, not settled findings. AI capabilities are current through 7 September 2026.*